UNIT

FOUR

Straight lines and planes in space

Exercise **12** Equation of a straight line in space.

Exercise **13** Equation of a plane in space.

Exercise **14** Relative positions of two planes in space.



Exercise 12

On equation of a straight line in space

From the school book | Understand | Apply | Higher Order Thinking Skills

First Exercise on the equation of the straight line in space

1 **Find a direction vector for each of the following straight lines :**

(1) The straight line passing through the origin point and point $(-1, 2, -2)$

(2) The straight line passing through the points C $(0, -2, 3)$ and D $(1, 1, -1)$

2 **Find the direction cosines of the straight line if its direction ratios :**

(1) $-1, 2, 3$ (2) $1, 1, 1$

3 A straight line passing through the two points $(6, -7, -1)$, $(2, -3, 1)$ find the direction ratios and the direction cosines for this straight line.

4 Find the vector form of the equation of the straight line passing through the point $(4, -2, 5)$ and its direction vector is $(1, -2, 2)$, then find another point on this straight line.

5 Find the different forms of the straight line $\overleftrightarrow{AB}$, where A $(-1, 4, 2)$, B $(3, -2, 6)$

6 Find all the different forms of the equation of the straight line $\frac{x+4}{3} = \frac{2y+5}{2} = \frac{4-z}{4}$, then find a point belongs to the straight line.

7 Find the equation of the straight line passing through the two points A $(3, 2, 4)$, B $(5, 8, 0)$, then prove that A , B , C are collinear where C $(4, 5, 2)$

8 **Choose the correct answer from those given :**

(1) Which of the following gives the direction cosines of a straight line ?

(a) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ (b) $(1, 1, 1)$

(c) $(1, -1, -1)$ (d) $(1, -1, 1)$

(2) The direction cosines of the straight line passing through the two points $(4, 3, -5)$, $(-2, 1, -8)$ could be .....

(a) $(6, 2, 3)$ (b) $(2, 4, -13)$ (c) $\left(\frac{6}{7}, \frac{2}{7}, \frac{3}{7}\right)$ (d) $\left(\frac{1}{2}, \frac{-\sqrt{3}}{2}, 0\right)$

(3) The direction cosines of the $\mathcal{X}$-axis could be .....

(a) $(1 , 0 , 0)$ (b) $(0 , 1 , 0)$ (c) $(0 , 0 , 1)$ (d) $(1 , 1 , 1)$

(4) The straight line which makes a direction angles of measure 60° with the y-axis and 60° with the z-axis , then it makes a direction angle with $\mathcal{X}$-axis of measure .....

(a) 60° (b) 30° (c) 45° (d) 75°

(5) ***(1st Session 2017)*** If the direction cosines of a straight line whose direction ratios $(-1 , 2 , 3)$ could be .....

(a) $\left(\frac{-1}{4} , \frac{1}{2} , \frac{3}{4}\right)$ (b) $\left(\frac{1}{\sqrt{14}} , \frac{2}{\sqrt{14}} , \frac{3}{\sqrt{14}}\right)$

(c) $\left(\frac{-1}{\sqrt{14}} , \frac{2}{\sqrt{14}} , \frac{3}{\sqrt{14}}\right)$ (d) $\left(\frac{-1}{14} , \frac{1}{7} , \frac{3}{14}\right)$

(6) If the direction angles of a straight line are $\theta_{\mathcal{X}} , \theta_y , \theta_z$, then $\cos 2\,\theta_{\mathcal{X}} + \cos 2\,\theta_y + \cos 2\,\theta_z =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(7) If the direction angles of a straight line are $\theta_{\mathcal{X}} , \theta_y , \theta_z$, then $\sin^2\theta_{\mathcal{X}} + \sin^2\theta_y + \sin^2\theta_z =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(8) The direction cosines of the straight line perpendicular to the $\mathcal{X}$ z-plane could be .....

(a) $(1 , 0 , 0)$ (b) $(0 , 1 , 0)$ (c) $(0 , 0 , 1)$ (d) $(1 , 0 , 1)$

(9) If the direction cosines of a straight line is $\left(\frac{1}{c} , \frac{1}{c} , \frac{1}{c}\right)$, then .....

(a) $c > 0$ (b) $0 < c < 1$ (c) $c = \pm\sqrt{3}$ (d) $c < 2$

(10) The equation of the straight line passing through the point A $(-1 , 0 , 2)$ and the vector $\vec{d} = (1 , -1 , 3)$ is a direction vector of it is .....

(a) $\frac{\mathcal{X}+1}{3} = \frac{y}{1} = \frac{z-2}{-1}$ (b) $\frac{\mathcal{X}+1}{1} = \frac{y}{-1} = \frac{z-2}{3}$

(c) $\frac{\mathcal{X}-1}{3} = \frac{y}{-1} = \frac{z}{1}$ (d) $\frac{\mathcal{X}+2}{1} = \frac{y-1}{-1} = \frac{z}{2}$

(11) The equation of the straight line passing through the two points $(2 , -1 , 3)$, $(0 , 3 , 1)$ is .....

(a) $\vec{r} = (2 , -1 , 3) + t\,(2 , -4 , 2)$ (b) $\vec{r} = (2 , -1 , 3) + t\,(2 , 2 , 4)$

(c) $\vec{r} = (2 , -4 , 2) + t\,(2 , -1 , 3)$ (d) $\vec{r} \,.\, (2 , -4 , 2) = 0$

(12) The equation of the straight line passing through the two points $(-2, 4, 2)$, $(7, -2, 5)$ is .....

(a) $\frac{x+2}{3} = \frac{y-4}{-2} = \frac{z-2}{1}$ (b) $\frac{x}{7} = \frac{y}{-2} = \frac{z}{5}$

(c) $\frac{x}{-2} = \frac{y}{4} = \frac{z}{2}$ (d) $\frac{x-1}{1} = \frac{y-2}{3} = \frac{z+1}{1}$

(13) ***(2nd Session 2022)*** The parametric equations of the straight line whose cartesian equation $y = 5, \frac{x-4}{3} = \frac{2z+1}{6}$ is .....

(a) $x = 3 + 4t$, $y = 5$, $z = 3 - \frac{1}{2}t$ (b) $x = 4 - 3t$, $y = 5$, $z = 3 + \frac{1}{2}t$

(c) $x = 4 + 3t$, $y = 5$, $z = 3 + \frac{1}{2}t$ (d) $x = 4 + 3t$, $y = 5$, $z = -\frac{1}{2} + 3t$

(14) The cartesian equatian of the straight line whose vector equation is $\vec{r} = (1, 3, 9) + t(5, 4, 2)$ is .....

(a) $x - 5 = \frac{y-4}{3} = \frac{z-2}{9}$ (b) $\frac{x-1}{5} = \frac{y-3}{4} = \frac{z-9}{2}$

(c) $x + 5 = \frac{y+4}{3} = \frac{z+2}{9}$ (d) $\frac{x+1}{5} = \frac{y+3}{4} = \frac{z-9}{2}$

(15) The vector equation of the straight line whose cartesian equation : $\frac{x+3}{2} = \frac{2y-1}{5} = \frac{3z+2}{4}$ is .....

(a) $\vec{r} = (-3, \frac{1}{2}, \frac{-2}{3}) + t(2, \frac{5}{2}, \frac{4}{3})$ (b) $\vec{r} = (-3, 1, -2) + t(2, 5, 4)$

(c) $\vec{r} = (2, 5, 4) + t(-3, 1, -2)$ (d) $\vec{r} = (2, \frac{5}{2}, \frac{4}{3}) + t(-3, \frac{-1}{2}, \frac{-2}{3})$

(16) ***(1st Session 2022)*** The straight line which passes through the point $(2, 2, 8)$ and its direction vector is $(3, 1, 4)$ cuts x-axis at the point .....

(a) $(-4, 0, 0)$ (b) $(4, 0, 0)$ (c) $(1, 0, 0)$ (d) $(-1, 0, 0)$

(17) The equation of the y-axis in space is .....

(a) $x = 0, z = 0$ (b) $z = 0, y = 0$ (c) $x = 0$ (d) $z = 0$

(18) The point lying on the straight line $\vec{r} = (2, -1, 3) + t(1, 2, -1)$ is .....

(a) $(1, 1, 1)$ (b) $(0, 2, -2)$ (c) $(3, 1, 2)$ (d) $(4, -3, 0)$

(19) Which point lies on the straight line $\frac{x}{3} = \frac{y+1}{1} = \frac{z-3}{2}$ such that its x-coordinate is twice its y-coordinate ?

(a) $(-6, -3, -1)$ (b) $(4, 2, -1)$ (c) $(6, 3, -1)$ (d) $(2, 1, -1)$

(20) If the three points $(5 , 2 , 4) , (6 , -1 , 2) , (8 , -7 , k)$ are belong to the same straight line , then k = ...

(a) 2 (b) -1 (c) 3 (d) -2

(21) The equation of the straight line passing through the point $(2 , 3 , -5)$ and makes equal angles with the coordinates axes is .....

(a) $\frac{x}{2} = \frac{y}{3} = \frac{z}{-5}$ (b) $x - 2 = y - 3 = z + 5$

(c) $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{-5}$ (d) $\frac{x-2}{\sqrt{3}} = \frac{y-3}{\sqrt{3}} = \frac{z+5}{3}$

(22) The equation of the straight line passes through the origin and makes equal angles with the coordinate axes is all the following except .....

(a) $x = y = z$ (b) $\frac{x}{\sqrt{2}} = \frac{y}{\sqrt{2}} = \frac{z}{\sqrt{2}}$

(c) $\frac{x-4}{2} = \frac{y-4}{2} = \frac{z-4}{2}$ (d) $\frac{x-1}{\sqrt{3}} = \frac{1-y}{\sqrt{3}} = \frac{z-1}{\sqrt{3}}$

(23) If the point $(2 , 3 , -3)$ moves parallel to the x-axis , then its path is a straight line whose equation is .....

(a) $\frac{x}{2} = \frac{y}{3} = \frac{z}{-3}$ (b) $y = 3 , z = -3$

(c) $2x + 3y - 3z = 0$ (d) $x = 0 , 3y - 3z = 0$

(24) The point (x , y , z) moves parallel to x-axis , which of the variables x , y , z is keep constant ?

(a) z , x (b) x , y (c) y , z (d) x

(25) The direction vector of the straight line : $x = \frac{y+5}{-2} = \frac{z+1}{3}$ is .....

(a) $(4 , 5 , 6)$ (b) $(3 , -5 , -1)$ (c) $(-3 , 5 , 1)$ (d) $(1 , -2 , 3)$

(26) Which of the following represents direction vector of the y-axis ?

(a) $(1 , 0 , 1)$ (b) $(0 , -1 , 0)$ (c) $(2 , 0 , 0)$ (d) $(1 , 1 , 1)$

(27) A straight line passes through the origin point and the point $(3 , -1 , 2)$, then $\cos \theta_z$ = ... where θ_z is measure of the angle which the straight line makes with the positive direction of z-axis

(a) $\frac{1}{2}$ (b) $\frac{-1}{\sqrt{14}}$ (c) $\frac{2}{\sqrt{14}}$ (d) $\sqrt{14}$

(28) A direction vector of the straight line $\frac{2x-3}{4} = \frac{y+5}{7} = \frac{3z-8}{6}$ is .....

(a) $(4 , 7 , 6)$ (b) $\left(\frac{3}{2} , -5 , \frac{8}{3}\right)$ (c) $(2 , 7 , 2)$ (d) $(3 , 4 , -2)$

(29) The vector direction of the straight line L : $\frac{x-2}{3} = \frac{y+3}{2}$, z = 4 equals .....

(a) (3 , 2 , 4) (b) (3 , 2 , 0) (c) (4 , 2 , 3) (d) (2 , 3 , 4)

(30) The direction vector of the straight line $2x + 3y = 5$, z = 4 is .....

(a) (2 , 3 , 4) (b) (3 , 2 , 4) (c) (3 , – 2 , 0) (d) (– 1 , 1 , 0)

(31) If ℓ , m , n are the direction cosines of a straight line in space , then .....

(a) $\ell = m = n = 1$ (b) $\ell + m + n = 1$

(c) $\ell^2 + m^2 + n^2 =$ zero (d) $\ell^2 + m^2 = 1 - n^2$

(32) ***(Speciman 2025)*** The equation of the straight line which passes through the point (2 , 3 , 5) and parallel to the straight line $\vec{r} = (-1 , 1 , 1) + k(5 , -4 , 7)$ is .....

(a) $\frac{x-2}{5} = \frac{y-3}{4} = \frac{z-5}{7}$ (b) $\frac{x+2}{5} = \frac{y-3}{-4} = \frac{z-5}{7}$

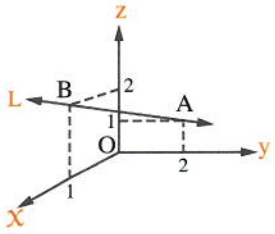
(c) $\frac{x+2}{5} = \frac{3-y}{4} = \frac{5-z}{7}$ (d) $\frac{x-2}{5} = \frac{3-y}{4} = \frac{z-5}{7}$

(33) ***(Speciman 2025)*** The equation of the straight line which passes through the point (1 , 2 , – 3) and parallel to y-axis is .....

(a) $\vec{r} = (1 , 2 , -3) + k(1 , 0 , 0)$ (b) $\vec{r} = (1 , 2 , -3) + k(0 , 1 , 0)$

(c) $\vec{r} = (1 , 2 , -3) + k(0 , 0 , 1)$ (d) $\vec{r} = (1 , 2 , -3) + k(1 , 1 , 1)$

(34) In the opposite figure :



The equation of the straight line is .....

(a) $x = \frac{y-2}{2} = z - 1$ (b) $x = \frac{y+2}{2} = \frac{z-1}{2}$

(c) $\frac{x}{2} = \frac{y-2}{-2} = z - 1$ (d) $x = \frac{y-2}{-2} = z - 1$

(35) ***(1st Session 2021)*** ABC is a triangle , where A (1 , 2 , 4) , B (– 2 , 0 , 5) , C (1 , 4 , 0) , if M is the intersection point of its medians , then the equation of straight line $\overleftrightarrow{AM}$ is .....

(a) $\vec{r} = (1 , 2 , 4) + t(-1 , 4 , 5)$ (b) $\vec{r} = (1 , 2 , 4) + t(-1 , 1 , 1)$

(c) $\vec{r} = (1 , 2 , 4) + t(-1 , 0 , -1)$ (d) $\vec{r} = (0 , 2 , 3) + t(1 , 2 , 4)$

(36) By using different real values for t in the equation of the straight line $\vec{r} = (2 , 1 , -3) + t(1 , -2 , 4)$ you get .....

(a) different direction vectors to the same straight line.

(b) different points on the straight line.

(c) different direction vectors perpendicular to the straight line.

(d) different direction cosines to the same straight line.

(37) If the straight line $\vec{r} = (1 , 2 , 3) + t (-2 , 1 , -1)$ intersects the plane $z = 2$ at point A and the plane $x = 3$ at point B , then length of $\overline{AB}$ = ... length units.

(a) 4 (b) 5 (c) $2\sqrt{6}$ (d) $6\sqrt{2}$

(38) *(Speciman 2025)* The equation of straight line passing through the point $(2 , 1 , -3)$ and intercepts a part of length 4 units from the positive part of z-axis is .....

(a) $\vec{r} = (2 , 1 , -3) + k (0 , 0 , 4)$ (b) $\vec{r} = (0 , 0 , 4) + k (2 , 1 , -3)$

(c) $\vec{r} = (0 , 0 , 4) + k (2 , 1 , -7)$ (d) $\vec{r} = (2 , 1 , -3) + k \vec{z}$

(39) *(Speciman 2025)* The equation of the straight line passing through the midpoint of $\overline{AB}$ where A $(5 , 2 , -3)$, B $(-1 , 4 , 1)$ and parallel to the straight line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{3-z}{2}$ is .....

(a) $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z+1}{2}$ (b) $\frac{x-2}{2} = \frac{y+3}{3} = \frac{z+1}{-2}$

(c) $\frac{x-2}{2} = \frac{y+3}{3} = \frac{z-1}{2}$ (d) $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z+1}{-2}$

(40) *(2nd Session 2023)* The equation of the straight line passing through the point $(1 , -2 , 4)$ and parallel to the straight line : $x = 2 + 3t$, $y = 3 - t$, $z = -1 + 4t$ is .....

(a) $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-4}{-1}$ (b) $\frac{x+1}{3} = \frac{y-2}{-1} = \frac{z+4}{4}$

(c) $\frac{x-1}{3} = \frac{y+2}{-1} = \frac{z-4}{4}$ (d) $\frac{x-1}{-3} = \frac{y+2}{1} = \frac{z-4}{-1}$

(41) *(2nd Session 2021)* The vector equation of the straight line passing through the point A $(2 , -1 , 4)$ and parallel to the bisector of the angle between $\overrightarrow{Oy}$, $\overrightarrow{OZ}$ in the plane y z is .....

(a) $\vec{r} = (2 , -1 , 4) + t (0 , 1 , 1)$ (b) $\vec{r} = (2 , -1 , 4) + t (0 , -1 , 1)$

(c) $\vec{r} = (2 , -1 , 4) + t (-1 , 0 , 1)$ (d) $\vec{r} = (2 , -1 , 4) + t (1 , 0 , -1)$

(42) *(Speciman 2025)* If the point B is the image of the point A $(1 , 2 , 3)$ by reflection on the x y plane , C $(3 , 2 , 5)$, then the equation of the straight line $\overleftrightarrow{BC}$ is .....

(a) $x - 3 = \frac{z-5}{8}$, $y = 2$ (b) $\frac{x-3}{2} = \frac{z-5}{8}$, $y = 2$

(c) $x - 3 = \frac{z-5}{8}$, $y = -2$ (d) $\frac{x-3}{2} = \frac{z-5}{8}$, $y = -2$

(43) *(1st Session 2023)* If ABCD is a parallelogram where A $(2 , 3 , -1)$, B $(0 , 5 , 3)$, C $(-2 , -3 , 1)$, then the vector equation of the straight line $\overleftrightarrow{CD}$ is .....

(a) $\vec{r} = (-2 , -3 , 1) + t (-1 , 1 , 2)$ (b) $\vec{r} = (-2 , -3 , 1) + t (2 , 3 , -1)$

(c) $\vec{r} = (-2 , -3 , 1) + t (0 , 5 , 3)$ (d) $\vec{r} = (-2 , 2 , 4) + t (-2 , -3 , 1)$

(44) ***(Speciman 2025)*** The equation of the straight line which passes through the point $(1\ ,2\ ,-4)$ and perpendicular to the two straight line $\vec{r_1} = (8\ ,-9\ ,10) + t_1\ (3\ ,-16\ ,7)$, $\vec{r_2} = (15\ ,-29\ ,5) + t_2\ (3\ ,8\ ,-5)$ is .....

(a) $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+4}{-6}$ (b) $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$

(c) $\frac{x-1}{-2} = \frac{y-2}{-3} = \frac{z+4}{6}$ (d) $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{-6}$

9 **Find the different forms of the equation of the straight line :**

(1) Passes through the two points $(2\ ,-1\ ,5)$ and $(-3\ ,1\ ,4)$

(2) ***(Trial 2023)*** Passes through the point $(2\ ,-1\ ,3)$ and cuts six units from the positive part of x-axis.

(3) Passes through the origin point and the vector $(-2\ ,3\ ,1)$ is a direction vector of it.

(4) Passes through the point $(4\ ,-2\ ,5)$ and the vector $\vec{d} = (2\ ,1\ ,-1)$ is a direction vector of it.

(5) Passes through the point $(2\ ,0\ ,-3)$ and parallel to the vector $\vec{A} = (2\ ,-3\ ,0)$

(6) Passes through the point $(3\ ,2\ ,5)$ and makes equal angles with the positive directions of the coordinate axes.

(7) Passes through the point A $(1\ ,-1\ ,0)$ and parallel to the straight line passing through the two points B $(-3\ ,2\ ,1)$, C $(2\ ,1\ ,0)$, then show that the point D $(-14\ ,2\ ,3)$ belongs to the straight line.

(8) Passing through the point $(-1\ ,4\ ,2)$ and parallel to the straight line $\frac{x-2}{4} = \frac{y}{5} = \frac{z+2}{3}$

(9) Passing through the point $(1\ ,5\ ,-2)$ and parallel to the straight line $x = 3 + 4t\ , y = 7 - 3t\ , z = -6 + 21t$

(10) Whose vector form of its equation : $\vec{r} = (1\ ,3\ ,9) + t\ (5\ ,4\ ,2)$

(11) Whose cartesian equation : $x - 2 = \frac{y+4}{5} = \frac{7-z}{2}$

(12) Whose cartesian equation : $\frac{x+3}{2} = \frac{2y-1}{5} = \frac{3z+2}{4}$

(13) Passing through the point A $(1\ ,-2\ ,5)$ and parallel to the bisector of the second quadrant of the plane y z

10 One of the vertices of a cube lies on the origin and the three edges drawn from this vertex lying on the positive directions of the coordinate axes if the length of the edge of a cube 5 length unit , find the equation of the straight line which carries the main diagonal drawn from the origin point.

11 Find the projection of point A (0 , 9 , 6) on the straight line passing through the two points B (1 , 2 , 3) and C (7 , – 2 , 5)

12 What can you say about the straight line with direction vector $\vec{d} = (a , b , 0)$?

13 **Discover the error :**

(1) The sum of the squares of direction ratios for any straight line equals 1

(2) The direction cosines of the straight line passing through the two points $(x_1 , y_1 , z_1) , (x_2 , y_2 , z_2)$ is $(x_2 - x_1 , y_2 - y_1 , z_2 - z_1)$

Second Exercise on the angle between two straight lines in space

1 **Choose the correct answer from those given :**

(1) The measure of the angle between the two lines $\vec{r_1} = (-2 , 5 , -7) + k (-6 , 6 , 8)$, $\vec{r_2} = (1 , -2 , 3) + \hat{k} (4 , 12 , -6)$ equals .....

(a) zero (b) 30° (c) 60° (d) 90°

(2) *(2nd Session 2017)* The measure of the angle between the two the straight lines whose direction cosines are $\left(\frac{2}{3} , \frac{-2}{3} , \frac{1}{3}\right)$ and $\left(\frac{1}{\sqrt{2}} , \frac{1}{\sqrt{2}} , 0\right)$ equals .....

(a) 60° (b) 30° (c) 90° (d) 120°

(3) The measure of the angle between the two straight lines : $2x = 3y = -z$ and $6x = -y = -4z$ equals .....

(a) zero (b) 45° (c) 90° (d) 120°

(4) The measure of the angle between the two lines which their direction ratios are (1 , 1 , 2) and (1 , 1 , 4) equals .....

(a) $\cos^{-1}\left(\frac{-5\sqrt{3}}{9}\right)$ (b) $\cos^{-1}\left(\frac{5\sqrt{3}}{9}\right)$

(c) $\sin^{-1}\left(\frac{5\sqrt{3}}{9}\right)$ (d) $\cos^{-1}\left(\frac{\sqrt{3}}{9}\right)$

(5) The measure of the angle between the two straight lines $\frac{x-3}{2} = \frac{z+1}{-2}$, $y = 1$ and $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z+1}{-2}$ equals .....

(a) 15° (b) 30° (c) 45° (d) 60°

(6) The measure of the angle between the straight line $\frac{x-1}{\sqrt{2}} = \frac{y-\sqrt{2}}{1} = \frac{z+1}{1}$ with the positive direction of z-axis is .....

(a) 30° (b) 45° (c) 60° (d) 120°

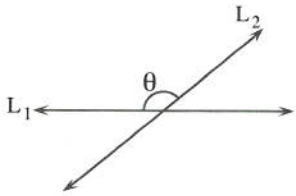
(7) If the measure of the angle between the two lines : $\frac{x}{a}=\frac{y}{2}=\frac{z}{1}$, $\frac{x}{2}=\frac{y}{1}=\frac{z}{-1}$ equals 60° , then the value of a = ...

(a) 1 or $\frac{2}{5}$ (b) 1 or $\frac{-13}{5}$ (c) $\frac{2}{5}$, $\frac{-13}{5}$ (d) $\pm\frac{13}{5}$

(8) 📖 **In the opposite figure :**

If $L_1 : x = 0$, $y = z$

$L_2 : y = 0$, $x = z$, then θ = ...



(a) 120° (b) 135°

(c) 150° (d) 165°

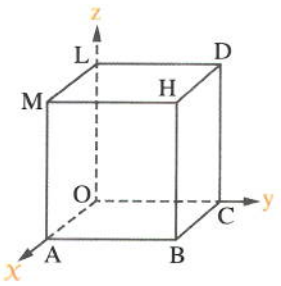
(9) Measure of the included angle between the two straight lines

$L_1 : x = 2y + 1$, $2y = 1 - z$ and $L_2 : 2x + y + z = 0$, $z + 2 = 0$ equals .....

(a) zero (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(10) ***(Speciman 2025)*** **In the opposite figure :**

If OABCDHML is a rectangular parallelepiped , and H (3 , 4 , 5) , then the measure of the angle between the two lines $\overleftrightarrow{OH}$, $\overleftrightarrow{LH}$ = ...°



(a) 37 (b) 45

(c) 60 (d) 90

2 📖 Find the measure of the angle between the two straight lines with direction cosines $\left(\frac{5}{13\sqrt{2}}, \frac{-12}{13\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(\frac{-3}{5\sqrt{2}}, \frac{4}{5\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ « 89° $\hat{7}$ »

3 📖 Find the measure of the angle between the two straight lines

$L_1 : 2x = 3y - 1 = z - 3$, $L_2 : \vec{r} = (2 , -1 , 5) + t(-1 , 1 , 2)$ « 50° $\hat{6}$ »

4 📖 Find the measure of the angle between the two straight lines :

$L_1 : \vec{r_1} = (2 , -1 , 3) + t_1 (-2 , 0 , 2)$ and $L_2 : x = 1$, $\frac{y-4}{3} = \frac{z+5}{-3}$ « 60° »

5 📖 Find the measure of the angle between the two straight lines

$L_1 : x = 2 - 5t$, $y = 1 - t$, $z = 3 + 4t$ and $L_2 : \frac{x+1}{3} = \frac{2-y}{4} = \frac{z}{2}$ « 85° $\hat{4}$ »

6 📖 **Find the measure of the angle between the two straight lines :**

(1) L_1 : passes through the two points (– 3 , 2 , 4) , (2 , 5 , – 2)

, L_2 : passes through the two points (1 , – 2 , 2) , (4 , 2 , 3)

(2) $L_1 : \vec{r} = (2 , -1 , 3) + t_1 (-1 , 4 , 2)$, $L_2 : \vec{r} = (0 , 2 , -1) + t_2 (1 , 1 , 3)$

(3) $L_1 : 2x = 3y = 4z$, $L_2 : \frac{x+1}{3} = \frac{y+2}{-2} = \frac{z}{-5}$ « $60° 31\hat{}$, $53° 41\hat{}$, $84° 2\hat{}$ »

7 Find the measure of the angle between the two straight lines :

$L_1 : x = 2 + 4t_1$, $y = 1 - t_1$, $z = 2 + 5t_1$

, $L_2 : x = -1 + t_2$, $y = 2t_2$, $z = 4 - t_2$ « $79° 6\hat{}$ »

8 Find the measure of the angle between the straight line $x = 1$, $y = 2$ and the straight line $y = -1$, $z = 0$ « $90°$ »

9 Prove that the angle between the two diagonals of the cube $= \cos^{-1}\left(\frac{1}{3}\right)$

10 Discover the error : If (a_1 , b_1 , c_1) and (a_2 , b_2 , c_2) are the direction ratios of the two straight lines L_1 and L_2 , then the measure of the angle between them is given by the relation $\cos\theta = |a_1 a_2 + b_1 b_2 + c_1 c_2|$

Third Exercise on positions of two straight lines in space

1 **Choose the correct answer from those given :**

(1) If $\vec{A} = (-2 , k , -3)$ is parallel to the straight line $\frac{x+2}{4} = \frac{y}{8} = \frac{z-1}{6}$, then k = ...

(a) -4 (b) -3 (c) -2 (d) -1

(2) If the straight line $\frac{x+3}{2} = \frac{y+1}{-6} = \frac{z-2}{k}$ is parallel to the straight line $\frac{x+2}{4} = \frac{y-5}{m} = \frac{z-1}{3}$, then k + m = ...

(a) $\frac{17}{2}$ (b) $\frac{11}{2}$ (c) $\frac{-15}{2}$ (d) $\frac{-21}{2}$

(3) *(2nd Session 2019)* If the straight line $\frac{x+2}{6} = \frac{y-1}{m} = \frac{z-1}{3}$ is perpendicular to the straight line $\frac{x-9}{-2} = \frac{y+8}{1}$, $z = 3$, then m = ...

(a) -12 (b) zero (c) 6 (d) 12

(4) If $L_1 : \frac{x+2}{-1} = \frac{y+3}{3} = \frac{z+5}{2}$ is perpendicular to $L_2 : \frac{x}{2} = \frac{y-5}{k} = \frac{z-6}{m}$, then 3 k + 2 m = ...

(a) -1 (b) 2 (c) m (d) 4

(5) *(1st Session 2024)* If the straight line passing through the two points $(5 , 1 , -2)$ and $(2 , 4 , 0)$ is perpendicular to the straight line whose equation : $\vec{r} = (1 , 2 , 3) + t(0 , m , 1)$, then m = ...

(a) $-\frac{3}{2}$ (b) $-\frac{2}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

(6) ***(1st Session 2019)*** If the straight lines $L_1 : x = 2t_1 - 1$, $y = t_1 + 1$, $z = t_1 - 1$, $L_2 : x = a t_2 - 1$, $y = 2t_2 + 1$, $z = b t_2 - 2$ are parallel , then $a + b =$ .....

(a) 4 (b) 2 (c) 6 (d) -2

(7) ***(1st Session 2021)*** If the two straight lines $L_1 : \vec{r} = t_1 (-2 , m , 7)$ and $L_2 : \frac{x-1}{n} = \frac{1-y}{4} = \frac{z-2}{2}$ are perpendicular , then : $n + 2m =$ .....

(a) 7 (b) -7 (c) 14 (d) -14

(8) If $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$, $\frac{x}{2} = \frac{y+2}{2} = \frac{z-3}{-2}$ are two straight lines , then they are : .....

(a) perpendicular. (b) coincident. (c) parallel. (d) intersecting.

(9) If the two straight lines : $\vec{r_1} = \vec{A_1} + t_1 \vec{d_1}$, $\vec{r_2} = \vec{A_2} + t_2 \vec{d_2}$ are parallel , then .....

(a) $\vec{d_1} = \ell \vec{d_2} , \ell \in \mathbb{R}^*$ (b) $\vec{d_2} // (\vec{A_2} - \vec{A_1})$

(c) $\vec{d_1} // (\vec{A_2} - \vec{A_1})$ (d) $\vec{d_1} . \vec{d_2} =$ zero

(10) If (a , b , c) and $(\grave{a} , \grave{b} , \grave{c})$ are two direction vectors of two perpendicular straight lines , then .....

(a) $a\grave{a} = b\grave{b} = c\grave{c} = 1$ (b) $a\grave{a} + b\grave{b} + c\grave{c} = 0$

(c) $\frac{a}{\grave{a}} = \frac{b}{\grave{b}} = \frac{c}{\grave{c}}$ (d) $a + b + c = \grave{a} + \grave{b} + \grave{c}$

(11) The two straight lines $\frac{x-1}{2} = \frac{y-2}{4} = \frac{z-3}{7}$ and $\frac{x-1}{4} = \frac{y-2}{5} = \frac{z-3}{7}$ are .....

(a) perpendicular. (b) intersecting. (c) skew. (d) parallel.

(12) The two straight lines : $x = y = z$ and $2x = 3y = z$ are .....

(a) skew. (b) perpendicular and intersecting.

(c) parallel. (d) intersecting and not perpendicular.

(13) If the two straight lines $L_1 : \vec{r_1} = (1 , 2 , 3) + t_1 (1 , 2 , 3)$ and $L_2 : \vec{r_2} = (-1 , 2 , 0) + t_2 (m , 2 , -1)$ are intersecting , then $m =$ .....

(a) $\frac{-8}{3}$ (b) $\frac{-5}{3}$ (c) zero (d) -3

(14) ***(1st Session 2024)*** The equation of the straight line passing through the point of intersection of the two straight lines : $L_1 : \frac{x-1}{2} = \frac{y+1}{3} = \frac{-z-1}{4}$, $L_2 : \vec{r} = (1 , -1 , -1) + t (-2 , 1 , 3)$ and parallel to the straight line : $\frac{2x+2}{4} = \frac{y-5}{3} = \frac{1-z}{-3}$ is .....

(a) $\vec{r} = (1 , -1 , -1) + t (2 , 3 , 3)$ (b) $\vec{r} = (2 , 3 , 3) + t (1 , -1 , -1)$

(c) $x = 1 + 3t , y = -1 + 2t , z = -1 - 3t$ (d) $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z-1}{3}$

2 Prove that the two straight lines : $L_1 : x = t_1 + 4$, $y = -4t_1 - 3$, $z = 7t_1 - 1$, $L_2 : x = 2t_2 + 1$, $y = -3t_2 - 1$, $z = 8t_2 - 10$ intersect at a point , then find their intersection point. « (5 , – 7 , 6) »

3 Prove that the two straight lines : $\frac{x-6}{2} = \frac{y-1}{2} = \frac{z-5}{3}$ and $\frac{x-2}{1} = \frac{y-3}{-2} = \frac{z+4}{3}$ are intersecting and find the coordinates of the intersection point. « (4 , – 1 , 2) »

4 *(2nd Session 2019)* Prove that the two straight lines : $\vec{r_1} = \hat{j} + t_1(\hat{i} + 2\hat{j} - \hat{k})$, $\vec{r_2} = (\hat{i} + \hat{j} + \hat{k}) + t_2(-2\hat{i} - 2\hat{j})$ intersect at a point , then find their intersection point. « (– 1 , – 1 , 1) »

5 Find the value of n which makes the two straight lines $L_1 : \vec{r_1} = (3 , -1 , n) + t_1(4 , 1 , 3)$, $L_2 : x = \frac{y-4}{-1} = \frac{z+1}{2}$ intersect at a point , then find the point of their intersection. « 7 , $(\frac{23}{5} , \frac{-3}{5} , \frac{41}{5})$ »

6 Find the value of a to make the two straight lines : $\frac{x+2}{2} = \frac{y}{-3} = \frac{z-1}{4}$, $\frac{x-3}{a} = \frac{y-1}{4} = \frac{z-7}{2}$ intersecting. « 3 »

7 Prove that the set of the equations : $\frac{x-1}{-6} = \frac{y-2}{-8} = \frac{z-3}{24}$, $\frac{x-4}{3} = \frac{y-6}{4} = \frac{z+9}{-12}$ represent the same straight line.

8 Find the equation of the straight line passing through the origin point and intersection point of the two straight lines.
$L_1 : \vec{r_1} = (2 , 3 , 4) + t(-4 , -4 , 5)$, $L_2 : x = -6$, $\frac{y+10}{1} = \frac{z+1}{3}$

9 Find the equation of the straight line passing through the point (2 , – 1 , 3) and intersection point of the two straight lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{3}$ and $\frac{x-3}{4} = \frac{y-5}{5} = \frac{z-6}{3}$

10 **State the necessary condition(s) to make the straight lines :**
$L_1 : x = x_1 + a_1 t_1$, $y = y_1 + b_1 t_1$, $z = z_1 + c_1 t_1$
$L_2 : x = x_2 + a_2 t_2$, $y = y_2 + b_2 t_2$, $z = z_2 + c_2 t_2$
(1) Parallel. **(2)** Perpendicular. **(3)** Intersect at a point.

11 If $\overrightarrow{OA} = \hat{i} - 2\hat{j} + \hat{k}$, $\overrightarrow{OB} = -\hat{i} - 3\hat{k}$, $\overrightarrow{OC} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\overrightarrow{OD} = 8\hat{i} + \hat{j} + 4\hat{k}$
Find the vector equation for each of the following straight lines :
(1) Passes through the two points A , B

(2) Passes through point D and parallel $\overleftrightarrow{BC}$

(3) Passes through point C and intersects $\overleftrightarrow{AB}$ orthogonally.

12 Find the equation of the straight line passing through the point $(2 , -1 , 3)$ and intersects the straight line $\vec{r}_1 = (1 , -1 , 2) + t\,(2 , 2 , -1)$ orthogonally.

13 Find the equation of the straight line passing through the origion point and intersects the straight line $\vec{r} = (3 , 1 , 4) + t\,(2 , 1 , 3)$ orthogonally.

14 Prove that the two straight lines :

$\vec{r}_1 = (3 , -3 , 5) + t_1\,(0 , -5 , 5)$, $\vec{r}_2 = (-2 , 3 , 1) + t_2\,(5 , -1 , -1)$

are perpendicular and intersect at a point , then find the coordinates of their intersection point.

15 *(1st Session 2019)* Prove that the two straight lines :

$\vec{r}_1 = (3 , -1 , 2) + t_1\,(4 , 1 , 3)$, $\vec{r}_2 = (0 , 4 , -1) + t_2\,(1 , -1 , 2)$ are skew.

16 Prove that the two lines :

$\vec{r}_1 = (1 , 2 , 4) + t_1\,(2 , -1 , 1)$, $\vec{r}_2 = (1 , 1 , 1) + t_2\,(-2 , 7 , 11)$

are orthogonal , then show that they are skew.

17 Prove that the two straight lines : $\frac{x+1}{-5} = \frac{y-4}{7} = z$ and $\frac{x}{2} = \frac{y}{1} = \frac{z}{3}$ are orthogonal , then determine if they are intersecting or skew.

18 Prove that the straight line passing through the two points $(2 , 3 , 4)$, $(3 , 4 , 5)$ is perpendicular to the straight line passing through the two points $(-2 , -3 , 6)$, $(4 , 0 , -3)$, then determine if they are intersecting or skew.

19 Find the value of n which makes the straight line passing through the two points A $(n , 1 , -1)$, B $(2 , 0 , 2n)$ perpendicular to the straight line passing through the two points C $(4 , 2n , 1)$, D $(2 , 3 , 2)$

« 1 »

Fourth Exercise on the distance between a point and a straight line in space

1 **Choose the correct answer from those given :**

(1) The distance between the point $(3 , 4 , 5)$ from the y-axis equals ..... length units.

(a) 3 (b) 5 (c) $\sqrt{34}$ (d) $\sqrt{41}$

(2) *(1st Session 2022)* The length of the perpendicular drawn from the point A $(0 , 3 , 0)$ to the straight line $\vec{r} = (3 , 2 , 1) + t\,(2 , -1 , 4)$ equals ..... length unit.

(a) $\sqrt{21}$ (b) $\sqrt{\frac{121}{21}}$ (c) $\sqrt{10}$ (d) $\sqrt{\frac{110}{21}}$

(3) *(2nd Session 2021)* The perpendicular distance between the point $(2 , 4 , 7)$ and the straight line $2x - 4 = \frac{2y - 8}{3} = \frac{2z - 14}{5}$ equals ..... length units.

(a) zero (b) 1 (c) 2 (d) 5

(4) The shortest distance between the point (a , b , c) and the x-axis equals .....

(a) $|c|$ (b) $|a|$ (c) $|b|$ (d) $\sqrt{b^2 + c^2}$

(5) The length of the perpendicular drawn from the point $(-1 , 0 , 1)$ to the straight line $\frac{x - 1}{2} = \frac{y - 1}{1} = \frac{z + 1}{-1}$ equals ..... length units.

(a) $\sqrt{30}$ (b) $\sqrt{6}$ (c) $\frac{\sqrt{30}}{6}$ (d) $2\sqrt{6}$

(6) If the length of the perpendicular drawn from the point $(2 , -1 , m)$ on the straight line $x - 1 = 3 - y = z$ equals 5 length unit , then number of possible values of m equals .....

(a) zero (b) 1 (c) 2 (d) infinite number.

(7) *(1st Session 2021)* If the perpendicular distance between the point $(-1 , 2 , m)$ and the straight line : $\vec{r} = (-1 , 3 , 0) + t(0 , -3 , 0)$ is 8 length unit , then the value of m equals ..... , where $m \in \mathbb{R}^+$

(a) 4 (b) 16 (c) 8 (d) 2

(8) If the point $(1 , -1 , 2)$ is the projection of the point $(0 , 3 , 1)$ on the straight line $\frac{x - a}{\ell} = \frac{y - 2}{3} = \frac{z - b}{4}$, then $a + b =$ .....

(a) 8 (b) 13 (c) 14 (d) 15

(9) *(2nd Session 2022)* The perpendicular distance between the two straight lines $L_1 : \vec{r} = (2 , -1 , 3) + t_1(-4 , -4 , 2)$ and $L_2 : \vec{r} = (1 , -1 , 2) + t_2(2 , 2 , -1)$ equals ..... length unit.

(a) $\frac{17}{3}$ (b) $\sqrt{\frac{17}{3}}$ (c) $\frac{\sqrt{17}}{3}$ (d) $3\sqrt{17}$

2 Find the length of the perpendicular drawn from point $(2 , 1 , -4)$ to the straight line $\vec{r} = (1 , -1 , 2) + t(2 , 3 , -2)$ « 4.18 length unit »

3 Find the distance between the point $(-1 , 2 , 5)$ and the straight line passing through the point $(3 , 4 , 5)$ and its direction ratios $(2 , -3 , 6)$ « 4.46 length unit »

4 Find the length of the projection of the vector passing through $(-1 , 2 , 0)$ and $(1 , -1 , 2)$ to the straight line $\vec{r} = (2 , 3 , -4) + t(3 , -3 , 2)$ « 4.1 length unit »

5 Find the perpendicular distance from point $(3 , -1 , 7)$ to the straight line passing through the two points $(2 , 2 , -1)$ and $(0 , 3 , 5)$ « 5.38 length unit »

6 Find the distance between the point $(-1 , 2 , 6)$ and the straight line passing through the point $(2 , 3 , -4)$ paralleling the vector $(6 , 3 , -4)$ « 7 length unit »

7 Find the distance between the point $(-2 , 3 , -4)$ and the straight line :

$$\frac{x+2}{3} = \frac{2y+3}{4} = \frac{3z+4}{5}$$

« 5.1 length unit »

8 Find the distance between the point $(1 , -2 , 3)$ and the straight line passing through the point $(2 , -3 , 5)$ and makes equal angles with the coordinate axes. « 2.2 length unit »

9 Find the distance between the two parallel straight lines :

$$L_1 : \frac{x-2}{2} = \frac{y+1}{1} = \frac{z}{3} \text{ and } L_2 : \frac{x-2}{2} = \frac{y-1}{1} = \frac{z-3}{3}$$

« 2.1 length unit »

10 Find the point A (x , y , z) which lies on the straight line : $x = 3 + t , y = 2 + t , z = 1 + t$ and is nearest to the point B $(1 , 2 , 1)$, then find the distance between B and the straight line by two different methods. « $\left(\frac{7}{3} , \frac{4}{3} , \frac{1}{3}\right)$, 1.6 length unit »

11 If $A = (1 , 2 , 4)$, $B = (-2 , 1 , 2)$, $C = (2 , 4 , -3)$, find the equation of the straight line $\overleftrightarrow{BC}$, then find the area of ΔABC « $\frac{15\sqrt{3}}{2}$ square unit »

12 Prove that the straight line passing through the point A $(4 , -3 , -1)$ and its direction ratios $(1 , -4 , 7)$ and the straight line passing through the point B $(1 , -1 , -10)$ and its direction ratios $(2 , -3 , 8)$ are intersected at a point as C , then find the area of ΔABC « 13.5 area unit »

13 Find the length of the perpendicular drawn from the vertex of a cube of side length ℓ to the diagonal which doesn't pass by it. « $\frac{\sqrt{6}}{3}\ell$ »

14 Find the smallest distance between the straight line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{4}$ and the x-axis. « 2.2 length unit »